

TABLE OF CONTENTS

M-210 CLOSED COOLING WATER SYSTEM

	Page No.
1. SCOPE OF WORK	M 210-2
2. CODES AND STANDARD	M 210-2
3. OPERATING AND DESIGN CONDITIONS	M 210-3
4. DETAILED REQUIREMENTS	M 210-3
4.1 Head Tank.....	M 210-3
4.2 Circulating Pumps.....	M 210-4
4.3 Heat Exchanger Design Requirements	M 210-4
4.4 Strainers	M 210-4
5. PERFORMANCE TESTING	M 210-5
5.1 Performance test.....	M 210-5
5.2 Shop test.....	M 210-5
6. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC).....	M 210-5
7. INSTRUMENTATIONS AND CONTROLS	M 210-5
8. ELECTRICAL SPECIFICATIONS.....	M 210-5
8.1 General.....	M 210-5
8.2 Motors.....	M 210-5

M 210 CLOSED COOLING WATER SYSTEM

1. SCOPE OF WORK

The closed cooling system for the steam cycle related plant and common auxiliaries of the Combined Cycle Plant shall be provided by the Bidder for each Block and shall include but not be limited to:

- (1) Two (2) closed cooling pumps (100% capacity each)
- (2) Two (2) closed cooling heat exchanger (100% capacity each).
- (3) Head Tank complete with automatic water level control valve.
- (4) Associated piping, valve, fittings and support
- (5) Soles plates, subsole plates and final grounding.
- (6) Foundations for all equipment.
- (7) All required electrical and I & C equipment.
- (8) All accessories and appurtenance required for a complete functional system.

2. CODES AND STANDARD

The Bidder shall comply with the requirements of **Part 4, Subsection 2.5 “Codes and Standards”**. The Codes and Standards of the following organizations extracted from **Part 4 Subsection 2.5** or international equivalent standards are specifically applicable to the design, manufacture and testing of the equipment included in this specification.

AISI	American Iron and Steel Institute
ANSI	American National Standards Institute
ASTM	American Society of Testing and Materials
BS	British Standard
DIN	German Institute for Standardization
EN	European Committee for Standardization
HIS	Hydraulic Institute Standards

IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronic Engineers
JIS	Japan Industrial Standard

3. OPERATING AND DESIGN CONDITIONS

The Closed Cooling Water (CCW) system shall be designed and installed to provide the cooling water to all equipment requiring fresh water cooling.

Treated demineralized water shall be passed in series through the equipment being served and through tubular heat exchangers. These heat exchangers shall reject heat from the CCW system to cooling water taken from the CW pump outlet line and discharged to the condenser CW outlet pipe.

The CCW supply and return lines for each plant item requiring cooling water shall be provided with isolating and flow balancing valves. Isolation valves shall not be used as flow balancing valves.

Individual return lines from all pump heat exchangers shall be equipped with flow indication devices. Plug type flow balancing valves shall be used to adjust flow to major plant users.

Provision shall be made in the system for manual dosing of chemicals to permit conditioning of the CCW system.

All piping, valves and supports shall be in accordance with **Part 4 Section 3, M-001 and M-004.**

4. DETAILED REQUIREMENTS

4.1 Head Tank

Each closed cooling system shall have a head tank of capacity to allow provision for make-up to the closed cooling system and to accommodate thermal expansion and contraction of the system water content.

The tank shall be of all welded construction, complete with pipe connections, manholes, lifting lugs and support feet.

Vessel seams shall be so positioned that they do not pass through vessel

connections. Inside seam weld shall be ground smooth for application of the internal protection coatings.

Each tank shall be internally lined with an epoxy paint to the approval of the Owner.

The outside surface shall be prepared and painted in accordance with **Part 4 Section 3, S-420, S-430**.

4.2 Circulating Pumps

Each unit system shall be provided with centrifugal pumps, to circulate water through the closed cooling water circuit.

All pumps shall be of the single stage, general purpose type. Pump operating speed shall be 1500 rpm.

The manufacture, inspection and testing of pumps shall comply with the appropriate standards of the Hydraulic Institute or governing specifications. For the detail requirements refer to **Part 4 Section 3, M-006 “Horizontal Centrifugal Pumps”**.

4.3 Heat Exchanger Design Requirements

The close cooling circuit shall be provided with two (2) x 100 % capacity heat exchangers (one duty, one standby) to transfer heat from the fresh water cooling to circulating water.

The tube side hydraulic characteristics of the heat exchangers shall be consistent with C.W system hydraulic characteristics as required by the main condenser.

The exchangers shall be furnish in accordance with **Part 4 Section 3, M-010, “Miscellaneous Heat Exchangers”**.

4.4 Strainers

The Bidder shall be provide self cleaning strainers at inlet of pumps. Strainer aperture shall not exceed 0.5 mm.

5. PERFORMANCE TESTING

5.1 Performance test

Bidder shall run a performance test after the equipment is put in service verify satisfactory performance.

5.2 Shop test

Refer to **Part 4, Section 3, M-006, M-010, M-014**

6. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Quality assurance shall conform to the requirement, of **Part 4, Subsection 2.7.**

7. INSTRUMENTATIONS AND CONTROLS

The Bidder shall be responsible for designing, supplying and installing instrumentation and control devices as specified herein.

Temperature indicators shall be provided at the outlet of all major components.

The Bidder shall also be responsible for the complete design, supply and installing of all local instrumentation and control systems, devices, test instruments and panels integral with the closed cooling system.

Pressure test points, complete with isolation valves, shall be provided across all system coolers, including the CCW heat exchangers (tube side only)

Supply header pressure and temperature shall be monitored alarmed in the main control room.

Temperature indicators shall be provided at all inlet and outlet connection of the CCW heat exchangers.

8. ELECTRICAL SPECIFICATIONS

8.1 General

For general electrical requirements, see **Part 4 Section 3, E-001.**

8.2 Motors

For detailed electrical requirements, see **Part 4 Section 3, E-210** for Low

Voltage Motors.

(Last of page M-210)